

SEMAPHORES

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Write a C program to implement the **Producer–Consumer problem** using semaphores. The program should simulate a bounded buffer where a producer produces items and a consumer consumes items.

The program must use the following variables:

- `mutex` → to ensure mutual exclusion
- `full` → to count filled slots
- `empty` → to count empty slots
- `x` → to track number of items

The user should be able to choose:

1. Producer
2. Consumer
3. Exit

Sample Input 1:

1
1
2
3

Sample Output 1:

----- Producer Consumer Problem -----

1. Producer
2. Consumer
3. Exit

Enter your choice: 1 Producer produces item 1

Enter your choice: 1 Producer produces item 2

Enter your choice: 2 Consumer consumes item 2

Enter your choice: 3 Exiting program... Final buffer items produced: 1

Sample Input 2:

2

1

3

Sample Output 2:

----- Producer Consumer Problem -----

Enter your choice: 2 Buffer is empty!! Consumer cannot consume.

Enter your choice: 1 Producer produces item 1

Enter your choice: 3 Exiting program... Final buffer items produced: 1

CODE:

// Producer Consumer

#include <stdio.h>

#include <stdlib.h>

int mutex = 1, full = 0, empty = 5, x = 0;

int wait(int s)

```
{  
  
    return --s;  
  
}
```

```
int signal(int s)  
  
{  
  
    return ++s;  
  
}
```

```
void producer()  
  
{  
  
    mutex = wait(mutex);  
  
  
    empty = wait(empty);  
  
    full = signal(full);  
  
  
    x++;  
  
    printf("\nProducer produces item %d", x);  
  
  
    mutex = signal(mutex);  
  
}
```

```
void consumer()  
  
{  
  
    mutex = wait(mutex);  
  
  
    full = wait(full);  
  
    empty = signal(empty);  
  
  
    printf("\nConsumer consumes item %d", x);  
  
    x--;  
  
  
    mutex = signal(mutex);  
  
}  
  
  
int main()  
  
{  
  
    int n;  
  
  
  
    printf("\n----- Producer Consumer Problem -----");  
  
    printf("\n1. Producer");  
  
    printf("\n2. Consumer");  
  
    printf("\n3. Exit");
```

```
while(1)
{
    printf("\nEnter your choice: ");
    scanf("%d",&n);

    switch(n)
    {
        case 1:
            if(mutex==1 && empty!=0)
            {
                producer();
            }
            else
            {
                printf("\nBuffer is full!! Producer cannot produce.");
            }
            break;

        case 2:
            if(mutex==1 && full!=0)
            {
```

```
        consumer();
    }
    else
    {
        printf("\nBuffer is empty!! Consumer cannot consume.");
    }
    break;

case 3:

    printf("\nExiting program...");

    printf("\nFinal buffer items produced: %d\n", x);

    exit(0);

default:

    printf("\nInvalid choice. Try again.");

}

}

return 0;

}
```